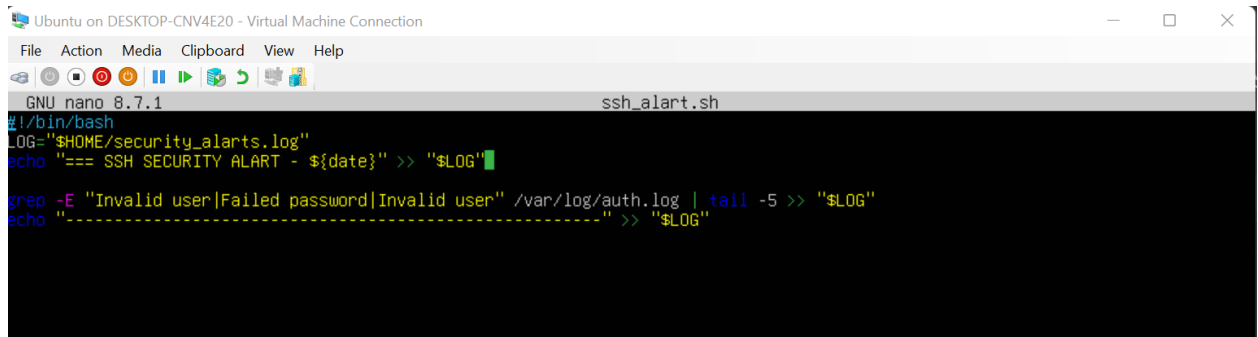


- Creation of the `ssh_alert.sh` script.



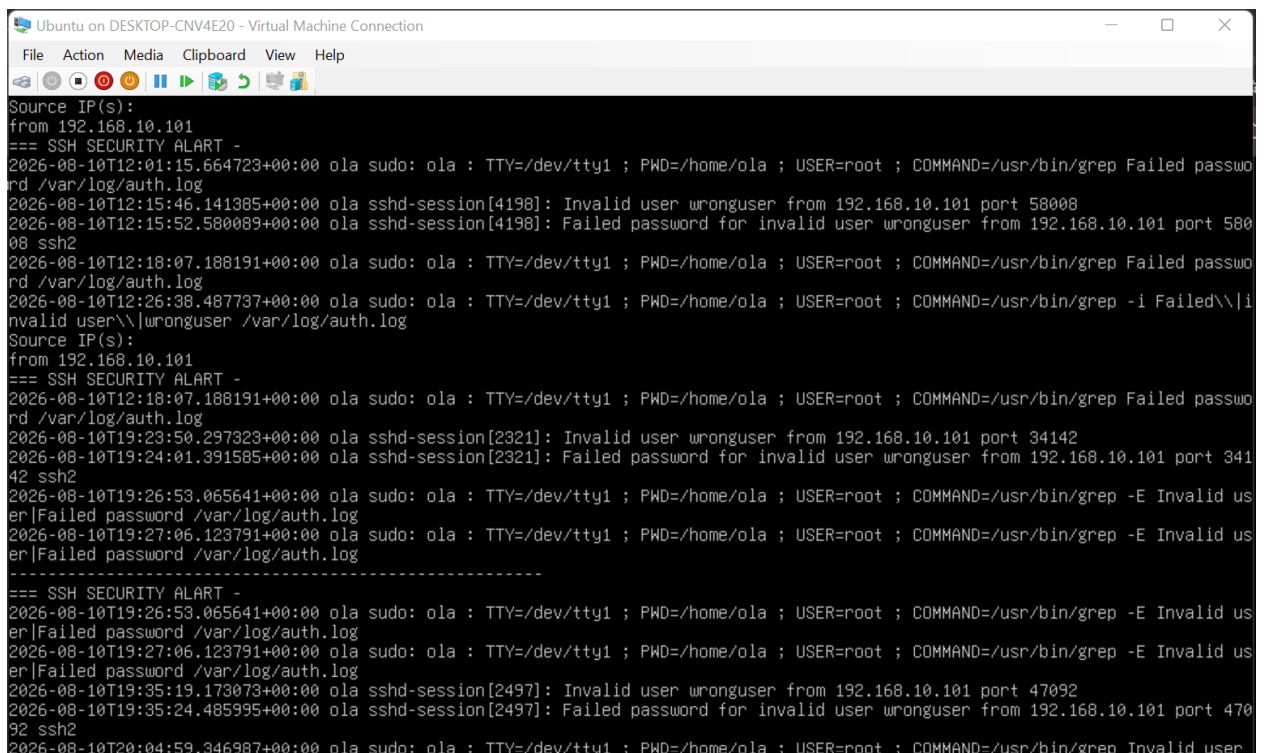
```

GNU nano 8.7.1 ssh_alert.sh
#!/bin/bash
LOG="$HOME/security_alerts.log"
echo "=== SSH SECURITY ALERT - ${date}" >> "$LOG"

grep -E "Invalid user|Failed password|Invalid user" /var/log/auth.log | tail -5 >> "$LOG"
echo "-----" >> "$LOG"

```

- Script execution.



```

Source IP(s):
from 192.168.10.101
=== SSH SECURITY ALERT -
2026-08-10T12:01:15.664723+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed passwo
rd /var/log/auth.log
2026-08-10T12:15:46.141385+00:00 ola sshd-session[4198]: Invalid user wronguser from 192.168.10.101 port 58008
2026-08-10T12:15:52.580089+00:00 ola sshd-session[4198]: Failed password for invalid user wronguser from 192.168.10.101 port 580
08 ssh2
2026-08-10T12:18:07.188191+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed passwo
rd /var/log/auth.log
2026-08-10T12:26:38.487737+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -i Failed\\i
nvalid user\\|wronguser /var/log/auth.log
Source IP(s):
from 192.168.10.101
=== SSH SECURITY ALERT -
2026-08-10T12:18:07.188191+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed passwo
rd /var/log/auth.log
2026-08-10T19:23:50.297323+00:00 ola sshd-session[2321]: Invalid user wronguser from 192.168.10.101 port 34142
2026-08-10T19:24:01.391585+00:00 ola sshd-session[2321]: Failed password for invalid user wronguser from 192.168.10.101 port 341
42 ssh2
2026-08-10T19:26:53.065641+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid us
er|Failed password /var/log/auth.log
2026-08-10T19:27:06.123791+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid us
er|Failed password /var/log/auth.log
-----
=== SSH SECURITY ALERT -
2026-08-10T19:26:53.065641+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid us
er|Failed password /var/log/auth.log
2026-08-10T19:27:06.123791+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid us
er|Failed password /var/log/auth.log
2026-08-10T19:35:19.173073+00:00 ola sshd-session[2497]: Invalid user wronguser from 192.168.10.101 port 47092
2026-08-10T19:35:24.485995+00:00 ola sshd-session[2497]: Failed password for invalid user wronguser from 192.168.10.101 port 470
92 ssh2
2026-08-10T20:04:59.346987+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Invalid user

```

- Detection results.

```

Ubuntu on DESKTOP-CNV4E20 - Virtual Machine Connection
File Action Media Clipboard View Help

Source IP(s):
from 192.168.10.101
=== SSH SECURITY ALERT -
2026-08-10T12:01:15.664723+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-08-10T12:15:46.141385+00:00 ola sshd-session[4198]: Invalid user wronguser from 192.168.10.101 port 58008
2026-08-10T12:15:52.580089+00:00 ola sshd-session[4198]: Failed password for invalid user wronguser from 192.168.10.101 port 58008 ssh2
2026-08-10T12:18:07.188191+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-08-10T12:26:38.487737+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -i Failed\\invalid user\\wronguser /var/log/auth.log
Source IP(s):
from 192.168.10.101
=== SSH SECURITY ALERT -
2026-08-10T12:18:07.188191+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-08-10T19:23:50.297323+00:00 ola sshd-session[2321]: Invalid user wronguser from 192.168.10.101 port 34142
2026-08-10T19:24:01.391585+00:00 ola sshd-session[2321]: Failed password for invalid user wronguser from 192.168.10.101 port 34142 ssh2
2026-08-10T19:26:53.065641+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
2026-08-10T19:27:06.123791+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
-----
=== SSH SECURITY ALERT -
2026-08-10T19:26:53.065641+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
2026-08-10T19:27:06.123791+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
2026-08-10T19:35:19.173073+00:00 ola sshd-session[2497]: Invalid user wronguser from 192.168.10.101 port 47092
2026-08-10T19:35:24.485995+00:00 ola sshd-session[2497]: Failed password for invalid user wronguser from 192.168.10.101 port 47092 ssh2
2026-08-10T20:04:59.346987+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Invalid user

```

- security_alerts.log containing detected SSH events.

```

Ubuntu on DESKTOP-CNV4E20 - Virtual Machine Connection
File Action Media Clipboard View Help

Source IP(s):
from 192.168.10.101
=== SSH SECURITY ALERT -
2026-08-10T12:01:15.664723+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-08-10T12:15:46.141385+00:00 ola sshd-session[4198]: Invalid user wronguser from 192.168.10.101 port 58008
2026-08-10T12:15:52.580089+00:00 ola sshd-session[4198]: Failed password for invalid user wronguser from 192.168.10.101 port 58008 ssh2
2026-08-10T12:18:07.188191+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-08-10T12:26:38.487737+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -i Failed\\invalid user\\wronguser /var/log/auth.log
Source IP(s):
from 192.168.10.101
=== SSH SECURITY ALERT -
2026-08-10T12:18:07.188191+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Failed password /var/log/auth.log
2026-08-10T19:23:50.297323+00:00 ola sshd-session[2321]: Invalid user wronguser from 192.168.10.101 port 34142
2026-08-10T19:24:01.391585+00:00 ola sshd-session[2321]: Failed password for invalid user wronguser from 192.168.10.101 port 34142 ssh2
2026-08-10T19:26:53.065641+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
2026-08-10T19:27:06.123791+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
-----
=== SSH SECURITY ALERT -
2026-08-10T19:26:53.065641+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
2026-08-10T19:27:06.123791+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep -E Invalid user|Failed password /var/log/auth.log
2026-08-10T19:35:19.173073+00:00 ola sshd-session[2497]: Invalid user wronguser from 192.168.10.101 port 47092
2026-08-10T19:35:24.485995+00:00 ola sshd-session[2497]: Failed password for invalid user wronguser from 192.168.10.101 port 47092 ssh2
2026-08-10T20:04:59.346987+00:00 ola sudo: ola : TTY=/dev/tty1 ; PWD=/home/ola ; USER=root ; COMMAND=/usr/bin/grep Invalid user

```

- Source IP identified from the detection output.

